Disaster Recovery

Part 1:

First 3 Actions:

1: The first thing to check for is any physical hazards like if anything fell on cables or and that all the hardware is in working condition or if there is a power outage.

2: The next thing I would do I check the power supply units to check whether it was a hardware failure or a power outage.

3: The last thing I would do is notify higher management about the outage and let them know what the financial impact may be and the severity of the problem (that I hopefully was able to figure out).

Top Priority:

My top priority would be the well-being of those in the workplace. Shortly behind that would be the confidentiality of the information and notifying management of the financial impact and how to minimize the monetary loss.

Communication:

The first people I would communicate with would be management, let them know the severity of the situation and estimated down time and recovery. Next would be the employees, and the users, to inform them of the issue and give them a downtime estimate. Next would be IT and Vendors to organize hardware replacements. Finally, facilities management to address the power outage.

Part 2:

Recovery Time Objective: The maximum amount of time that a company can afford to be offline before damage becomes too great.

Recover Point Objective: The maximum amount of data that a company can lose after some kind of outage or attack.

This scenario: Outage at 10:15 AM Monday and the most recent backup was 6 PM the previous day. The Current RPO is 16 hours and 15 minutes. Any kind of transactions, interactions or modifications that happened after 6pm have potential to be permanently lost.

Part 3:

Restore Services:

Step 1: Restore physical power and control the main distribution frame.

Step 2: Reboot or replace the failed switch and make sure VLAN's are still protected

Step 3: Restore the file server and verify data integrity against the most recent backups.

Step 4: Restart the email servers, collab tools and the internet firewall.

Step 5: Make sure the system is running smoothly and notify users.

Hot Site:

A hot site features real-time data replication and redundant infrastructure that can take over nearly instantaneously, drastically reducing downtime and paying for itself during a major disaster. A cold site would take too long and lead to losing too much money.

Systems to come back online:

1. Power and Environmental Infrastructure
2. Network Infrastructure
3. Identity Management
4. File servers and databases
5. Internet access

Reducing Downtime in the future:

Moving towards continuous data protection and more frequent automated snapshot backups instead of a single daily snapshot. Installing uninterruptible power supplies with automatic generator transfer switches.

Part 4:

Active-active:

Two web servers sit behind a load balancer, and both actively prove live user traffic. If one of the servers fail, the load balancer routes all the traffic to the remaining server immediately preventing any service interruption.

Active passive:

A primary server handles all live interactions while another server stays powered on in sync while idle. If the main server crashes, the software activates the idle server and that becomes the main server for traffic.

Recommendation:

I would recommend active passive for this company; it offers sturdily high availability while also being simple and secure. Since it's simple it is also cost effective and eliminates single points of failure.

What tools prevent downtime:

Industrial-grade Uninterruptible Power Supplies

Strong backup generators

Dual homes core switches with redundant power supplies

Automated failover clustering software

Reflection:

The biggest risk in this scenario is the single point of failure. If the main core switch goes out, then everything it's connected to also goes out.

Only taking back up snapshots of data once a day is a mistake that made the outage worse. If backup snapshots were taken more often, not as much data would be lost.

Moving forward, upgrading the main distribution frame power infrastructure with industrial uninterruptable power supplies and backup generators. Adding a secondary redundant core switch configured in high availability would eliminate the single point of failure. Moving toward continuous data snapshots instead of a single daily backup would also help prevent data loss in the future.